

Re: 从零开始的布朗运动

李秋满

1 布朗运动的产生

随机游走: ξ_i i.i.d., $S_n = \sum_{i=1}^n \xi_i$, 其中

$$\mathbb{E}\xi_i = 0, \quad \mathbb{E}\xi_i^2 = \sigma^2.$$

在连续时间下 $S_t = S_{\lfloor t \rfloor}$ 。

一般时间间隔不一定是 1: $S_{\lfloor nt \rfloor} \rightarrow \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$ 。

观测时间缩小, 步长也相应减小。

由中心极限定理, 可理解 $\frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$ 有极限 $\rightarrow B_t, n \rightarrow \infty$ 。

极限的性质

- ① 独立增量性
- ② 平稳增量性

定义 1. 有下面性质的随机过程是布朗运动:

- ① 独立增量性
- ② 平稳增量性
- ③ $B(t+s) - B(s) \sim N(0, t), t \mapsto B_t$ 是连续的

定理 2. $B(t)$ 是布朗运动且 $B(0) = 0$ 当且仅当

- (1) $B(t)$ 是高斯过程
- (2) $\mathbb{E}B(t) = 0, \mathbb{E}B(t)B(s) = t \wedge s$
- (3) $t \mapsto B(t)$ 连续

证明. $\implies$: $B(t_n) - B(t_{n-1}), B(t_{n-1}) - B(t_{n-2}), \dots, B(t_1) - B(t_0), B(t_0)$ 独立, 且 $B(t_0) = 0, B(t_i) - B(t_{i-1}) \sim N(0, t_i - t_{i-1})$ 。

$$(B(t_n), B(t_{n-1}), \dots, B(t_1)) = (X_n, X_{n-1}, \dots, X_1) \begin{pmatrix} 1 & 1 & \dots & 1 \\ & 1 & \dots & 1 \\ & & \ddots & \vdots \\ & & & 1 \end{pmatrix}$$

故 $B(t_n), B(t_{n-1}), \dots, B(t_1)$ 联合高斯, 故为高斯过程。

已知 $B(t) \sim N(0, t) \Rightarrow \mathbb{E}B(t) = 0$, 若 $t > 0$,

$$\begin{aligned}\mathbb{E}(B(t)B(s)) &= \mathbb{E}(B(t)(B(s) - B(t) + B(t))) \\ &= \mathbb{E}B(t)^2 + \mathbb{E}B(t)\mathbb{E}(B(s) - B(t)) = t.\end{aligned}$$

连续性显然。

$\Leftarrow$: $B(0) \sim N(\mu, \sigma^2)$, $\mathbb{E}B(0) = 0$, $\mathbb{E}B(0)^2 = 0 \Rightarrow B(0) \sim N(0, 0)$, $\Rightarrow B(0) = 0$ a.s.。

$B(t_1), B(t_2), \dots, B(t_n)$ 是联合高斯的, 所以

$$B(t_1), B(t_2) - B(t_1), B(t_3) - B(t_2), \dots, B(t_n) - B(t_{n-1})$$

是联合高斯分布。

$$\mathbb{E}[(B(t_i) - B(t_{i-1}))(B(t_j) - B(t_{j-1}))] = t_i - t_j - t_{i-1} + t_{j-1} = 0 \quad (t_i > t_{i-1} \geq t_j > t_{j-1}).$$

不相关 $\Rightarrow$ 独立, 对多个同理, 故上面是独立增量的。

$$\mathbb{E}(B(t_n) - B(t_{n-1})) = 0, \quad \mathbb{E}(B(t_n) - B(t_{n-1}))^2 = t_n - t_{n-1},$$

$$B(t_n) - B(t_{n-1}) \sim N(0, t_n - t_{n-1}).$$

□

重对数律

$$\limsup_t \frac{B_t}{\sqrt{2t \log \log t}} = 1 \quad \text{a.s.}, \quad \liminf_t \frac{B_t}{\sqrt{2t \log \log t}} = -1 \quad \text{a.s.}$$

时间反转

$tB_{1/t}$ 也是标准布朗运动。

证明. 1. $B_{1/t_1}, B_{1/t_2}, \dots, B_{1/t_n}$ 是高斯的 $\Rightarrow tB_{1/t}$ 是高斯过程。

2. $\mathbb{E}tB_{1/t} = t\mathbb{E}B_{1/t} = 0$,

$$\mathbb{E}[tB_{1/t} \cdot sB_{1/s}] = ts \left(\frac{1}{t} \wedge \frac{1}{s} \right) = s \wedge t.$$

3. $t \mapsto tB_{1/t}$ 连续: 若 $t \neq 0$, $t \mapsto t$, $t \mapsto B_{1/t}$ 连续 $\Rightarrow tB_{1/t}$ 连续; $t = 0$ 验证 $\lim_{t \rightarrow 0} tB_{1/t} = 0$,

即 $\lim_{t \rightarrow \infty} \frac{B_t}{t} = 0$.

□

由重对数律

$$\limsup_t \frac{B_t}{t} = \limsup_t \frac{B_t}{\sqrt{2t \log \log t}} \cdot \frac{\sqrt{2t \log \log t}}{t} = 0, \quad \liminf_t \frac{B_t}{t} = 0.$$

布朗运动的构造

$\{e_n\}$ 是 $L^2[0, 1]$ 中的标准正交基, ξ_n 是独立同分布的标准正态分布,

$$X_t \stackrel{L^2}{:=} \sum_n \langle \mathbf{1}_{[0,t]}, e_n \rangle \xi_n,$$

有 L^2 收敛的极限。

X_t 是一个中心化高斯过程。由于

$$\begin{aligned} \mathbb{E} X_t^2 &= \sum_n \langle \mathbf{1}_{[0,t]}, e_n \rangle^2 = \int \mathbf{1}_{[0,t]} dt = t, \\ \implies \mathbb{E} X_t X_s &= t \wedge s. \end{aligned}$$

对于连续性, 有下面定理:

定理 3. 一个过程 X_t $((\Omega, \mathcal{F}, \mathbb{P}))$, 若有 $p > 0$, $\beta > 0$ s.t.

$$\mathbb{E} |X_t - X_s|^p \leq C |t - s|^{1+\beta},$$

则有修正 $\tilde{X}$ locally Hölder continuous for every $\gamma \in (0, \frac{\beta}{p})$,

$$\mathbb{P} \left(\sup_{0 \leq s \neq t \leq 1} \frac{|\tilde{X}_t(\omega) - \tilde{X}_s(\omega)|}{(t-s)^\gamma} \leq \delta \right) = 1$$

即对某个随机变量 $\eta(\omega) > 0$, 某个 $\delta > 0$ 。

对于一个高斯过程 $B_t - B_s \sim N(0, t-s)$,

$$B_t - B_s = \sqrt{t-s} \xi,$$

$$\mathbb{E} |B_t - B_s|^p = C(p)(t-s)^{\frac{p}{2}} = C(p)(t-s)^{1+(\frac{p}{2}-1)}, \quad \text{取 } p > 2, \text{ 则 } \frac{p}{2} - 1 > 0.$$

取 p 足够大, $\gamma \in (0, \frac{1}{2})$, 则 $t \mapsto B_t$ 是 γ -Hölder 连续的。

高维布朗运动

由独立的一维布朗运动构成。

马氏可夫性

$$(\Omega, \mathcal{F}, \mathbb{P}) = (C([0, \infty), \mathbb{R}^n), \mathcal{G}, \mathbb{P}),$$

shift operator $\{\theta_t : t \geq 0\}$ 。

2 马氏性、反射原理与 Blumenthal 0-1 律

$$\theta_t : \Omega \longrightarrow \Omega, \quad w(\cdot) \longmapsto w(\cdot + t).$$

布朗运动在典则空间下 $B_t(w) = w(t)$, $t \geq 0$ 。

定义 $\mathcal{F}_t^\circ = \sigma(B_s, s \leq t)$, 右修正 $\mathcal{F}_t^+ = \bigcap_{s>t} \mathcal{F}_s^\circ$.

$$B_{t+s} - B_s = \lim_{r \downarrow s} B_r - B_s \quad \text{与 } \mathcal{F}_s^+ \text{ 无关.}$$

$\forall s \geq 0$, Y 为有界 $\mathcal{F}_\infty^\circ$ 可测函数, $f(B_t)$,

$$\mathbb{E}_x(Y \circ \theta_s \mid \mathcal{F}_s^+) = \mathbb{E}_{B_s} Y.$$

布朗运动具有强马氏性:

$$\mathbb{E}_x(Y \circ \theta_T \mid \mathcal{F}_T^+) = \mathbb{E}_{B_T} Y, \quad Y \text{ } \mathcal{F}_\infty^\circ \text{ bounded.}$$

T 为 $\mathcal{F}_t^+$ 停时, $\mathcal{F}_T^+ := \{\Lambda \in \mathcal{F}_\infty^\circ : \Lambda \cap \{T \leq t\} \in \mathcal{F}_t^+\}$,

$$\theta_{T(w)}(w) = w(T(w) + \cdot).$$

反射原理

B_t 是 BM, $B_0 = 0$, $\{-B_t, t \geq 0\} \stackrel{d}{=} \{B_t, t \geq 0\}$.

则 $\forall a > 0$, $T_a = \inf\{t \geq 0 : B_t \geq a\}$,

$$\mathbb{P}_0(T_a < t) = 2\mathbb{P}(B_t > a).$$

$$\begin{aligned} \mathbb{P}_0(T_a < t) &= \mathbb{P}_0(T_a < t, B_t > a) + \mathbb{P}(T_a < t, B_t < a) \\ &= 2\mathbb{P}_0(T_a < t, B_t > a) \\ &= 2\mathbb{P}_0(B_t > a). \end{aligned}$$

Blumenthal's 0-1 law

若 $A \in \mathcal{F}_0^+$, $\mathbb{P}_0(A) = 0$ 或 1 。

$$A = A \cap A = A \cap A \circ \theta_0,$$

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{E}\mathbf{1}_A = \mathbb{E}\mathbf{1}_A \mathbf{1}_{A \circ \theta_0} = \mathbb{E}(\mathbb{E}(\mathbf{1}_A \cdot \mathbf{1}_{A \circ \theta_0} \mid \mathcal{F}_0^+)) \\ &= \mathbb{E}(\mathbf{1}_A \mathbb{E}(\mathbf{1}_{A \circ \theta_0} \mid \mathcal{F}_0^+)) = \mathbb{P}(A)^2 \\ &\implies \mathbb{P}(A) = 1 \text{ 或 } 0. \end{aligned}$$

例 4. $T_0^+ = \inf\{t > 0 : B_t > 0\}$, $\mathbb{P}(T_0^+ = 0) = 1$ 。

$$\begin{aligned} \mathbb{P}(T_0^+ = 0) &= \mathbb{P}\left(\bigcap_n \{T_0^+ < \tfrac{1}{n}\}\right) = \lim_n \mathbb{P}(T_0^+ < \tfrac{1}{n}) \\ &= \lim_n 2\mathbb{P}(B_{1/n} > 0) = 1. \end{aligned}$$

因而布朗运动立刻进入正半轴, 也立刻击中 0。

3 Itô 积分

$$\int_0^t f(s) d\varphi(s) \longrightarrow \int_0^t H(s) dB(s) \quad \text{不是有限变差.}$$

随机积分之定义: H 是 $\mathcal{F}_t^+$ 适应过程。

- Λ : $[0, \infty) \times \Omega$ 上由右连续适应过程生成的 σ -代数
- π' : 左连续
- $\pi := \sigma\{\mathbf{1}_{(a,b]}, 0 \leq a < b\}$

关系: ① $\pi' \subset \pi$; ② 对布朗运动有 $\Lambda = \pi$ 。

对于简单过程, H_s 是 $\mathcal{F}_t^+$ 适应, 有 $0 = t_0 < t_1 < t_2 < \cdots < t_n < \infty$,

$$H_s = H_{t_k}, \quad s \in [t_k, t_{k+1}), \quad k = 1, 2, 3, \dots, n.$$

定义积分

$$M_t = \int_0^t H_s dB_s = \sum_{i=1}^n H_{t_{i-1}}(B_{t_i} - B_{t_{i-1}}) + H_{t_n}(B_t - B_{t_n}) \quad \text{与初值无关,}$$

若 $t_n \leq t < t_{n+1}$ 。

M_t 的性质

$$\mathbb{E}M_t = \sum_{i=1}^n \mathbb{E}(H_{t_{i-1}}(B_{t_i} - B_{t_{i-1}})) + \mathbb{E}(H_{t_n}(B_t - B_{t_n})),$$

$$\mathbb{E}(H_{t_i}(B_{t_{i+1}} - B_{t_i})) = \mathbb{E}\left(\mathbb{E}(H_{t_i}(B_{t_{i+1}} - B_{t_i}) \mid \mathcal{F}_{t_i}^+)\right) = \mathbb{E}\left(H_{t_i} \underbrace{\mathbb{E}(B_{t_{i+1}} - B_{t_i})}_{=0}\right) = 0.$$

$$\textcircled{1} \mathbb{E}M_t = 0$$

$$\textcircled{2} \mathbb{E}M_t^2 = \text{Var } M_t = \mathbb{E}\left(\sum_{i=1}^n H_{t_{i-1}}(B_{t_i} - B_{t_{i-1}})\right)^2$$

$$\begin{aligned} & \mathbb{E}\left(H_{t_i}(B_{t_{i+1}} - B_{t_i}) H_{t_j}(B_{t_{j+1}} - B_{t_j})\right), \quad t_i < t_j \leq \cdot \\ &= \mathbb{E}\left(H_{t_i} H_{t_j} \mathbb{E}((B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j}) \mid \mathcal{F}_{t_j}^+)\right) \\ &= \mathbb{E}\left(H_{t_i} H_{t_j} \mathbb{E}(B_{t_{i+1}} - B_{t_i} \mid \mathcal{F}_{t_j}^+) \mathbb{E}(B_{t_{j+1}} - B_{t_j} \mid \mathcal{F}_{t_j}^+)\right) \\ &= 0. \end{aligned}$$

$$\begin{aligned} \mathbb{E}M_t^2 &= \mathbb{E} \sum_i H_{t_{i-1}}^2 (B_{t_i} - B_{t_{i-1}})^2 = \mathbb{E} \sum_i H_{t_{i-1}}^2 (t_i - t_{i-1}) \implies \text{伊藤等距} \\ &= \mathbb{E} \int_0^t H_s^2 ds. \end{aligned}$$

$$\textcircled{3} t \mapsto M_t \text{ 连续, } M_t \text{ 是鞅}$$

$$\text{同一时段} \begin{cases} \text{连续性继承自布朗运动} \\ \mathbb{E}(M_t \mid \mathcal{F}_s^+) = M_s \end{cases} \implies \text{不同时段} \begin{cases} \text{拼接} \\ \text{检验性质} \end{cases}$$

一般积分

任何 L^2 适应过程 H , 在 $[0, T] \times \Omega$ 上, 有简单过程 H^n ,

$$\lim_n \mathbb{E} \int_0^T |H_s^n - H_s|^2 ds = 0.$$

定义

$$\int_0^T H_s dB_s = \lim_{n \rightarrow \infty} \int_0^T H_s^n dB_s \quad (L^2 \text{ 极限}).$$

同样有 $\mathbb{E}M_t = 0$, $\mathbb{E}M_t^2 = \mathbb{E} \int_0^t H_s^2 dB_s$, 鞅性。

连续性:

$$\begin{aligned} \mathbb{P} \left(\sup_{t \leq T} \left| \int_0^t H_s dB_s - \int_0^t H_s^n dB_s \right| \geq \lambda \right) &\leq \frac{1}{\lambda^2} \mathbb{E} \left| \int_0^t H_s dB_s - \int_0^t H_s^n dB_s \right|^2 \rightarrow 0 \\ \implies \int_0^t H_s^n dB_s &\implies \int_0^t H_s dB_s \implies t \mapsto \int_0^t H_s dB_s \text{ 连续}. \end{aligned}$$

性质: $B_t^2 - t$ 是鞅。

$$\begin{aligned} \mathbb{E}(B_t^2 - t - B_s^2 + s \mid \mathcal{F}_s^+) &= \mathbb{E}((B_t - B_s + B_s)^2 - t - B_s^2 + s \mid \mathcal{F}_s^+) \\ &= (t - s) + s - t + \mathbb{E}(2(B_t - B_s)B_s \mid \mathcal{F}_s^+) = 0. \end{aligned}$$

$\implies$ 二次变差过程。

二次变差

事实. (i) 给一个连续鞅 X_t , 存在唯一的连续适应过程 A_t , $A_0 = 0$, $X_t^2 - A_t$ 是局部鞅, A_t 记作 $\langle X \rangle_t$ 。

$$(ii) \sum_i (X_{t_{i+1}} - X_{t_i})^2 \xrightarrow{\mathbb{P}} \langle X \rangle_t, \quad \|\Pi\| \rightarrow 0.$$

所以对于 B_t , $\langle B \rangle_t = t$ 。

事实上 $X_{nk} = (t_{nk} - t_{n,k-1}) \xi_k^2$,

$$\varphi_k(t) = 1 + (t_{nk} - t_{n,k-1}) it + o(t),$$

$$\prod_k \varphi_k(t) = \prod_k \left(1 + (t_{nk} - t_{n,k-1}) it + o(t) \right) \rightarrow e^{itT}.$$

故 $\sum_k (B_{t_k} - B_{t_{k-1}})^2 \xrightarrow{\mathbb{P}} t$ 。

定理 5. 对于简单过程 H ,

$$\left\langle \int_0^t H_s dB_s \right\rangle_t = \int_0^t H_s^2 ds = \left(H^2 \cdot \langle B \rangle \right)_t.$$

进一步地

$$M_t = \left(\int_0^t H_s dB_s \right)^2 - \int_0^t H_s^2 ds$$

是鞅。

证明. 仅对同一时间段内进行证明。

$$\begin{aligned}
& \mathbb{E}(M_t | \mathcal{F}_s^+) - M_s \\
&= \mathbb{E} \left[\left(\sum_{i=1}^k H_{t_{i-1}}(B_{t_i} - B_{t_{i-1}}) + H_{t_k}(B_t - B_{t_k}) \right)^2 - \int_0^t H_s^2 ds - \left(\int_0^s H_u dB_u \right)^2 + \int_0^s H_u^2 du \mid \mathcal{F}_s^+ \right] \\
&= \mathbb{E} \left[2 \left(\int_0^s H_u dB_u \right) (H_{t_k}(B_t - B_{t_k})) + H_{t_k}^2 (B_t - B_{t_k})^2 - \int_s^t H_s^2 ds \mid \mathcal{F}_s^+ \right] \\
&= 0 + (t - s) - (t - s) = 0.
\end{aligned}$$

□

对于一般地 L^2 可积过程,

$$\left\langle \int_0^t H_s dB_s \right\rangle = \int_0^t H_s^2 ds = \int_0^t H_s^2 d\langle B \rangle_s.$$

证明.

$$M_t^n = \left(\int_0^t H_s^n dB_s \right)^2 - \int_0^t H_s^{2n} ds \text{ 是鞅} \quad \begin{matrix} \downarrow L^2 \\ \downarrow L^2 \end{matrix} \quad \int_0^t H_s dB_s, \int_0^t H_s^2 ds \implies \text{是鞅}.$$

□

关于鞅的二次变差的性质

① 停止过程: $\langle M^T \rangle_s = \langle M \rangle_{s \wedge T}$

② $\langle aM \rangle = a^2 \langle M \rangle$

$A_t = \langle M \rangle_t$, $M^2 - A$ 为鞅, $(aM)^2 - a^2 A$ 为鞅,

$(M_{t \wedge T})^2 - A_{t \wedge T} = (M^2 - A)_{t \wedge T}$ 是鞅.

② $\langle M \rangle = 0$, 则 $M_t \equiv M_0$ a.s.。

由于 M^2 是鞅 ($M_0 = 0$),

$$\mathbb{E}(M_t - M_0)^2 = \mathbb{E}M_t^2 - \mathbb{E}M_t M_0 + \mathbb{E}M_0^2 = 0.$$

协方差过程:

$$\langle X, Y \rangle := \frac{1}{2} (\langle X + Y \rangle - \langle X - Y \rangle).$$

事实. (i) $\langle X, Y \rangle$ 是唯一的有界变差适应过程 A_t , $A_0 = 0$, 且 $X_t Y_t - A_t$ 是局部鞅。

(ii) $\langle X, Y \rangle = \langle Y, X \rangle$

(iii) $\langle X + Y, Z \rangle = \langle X, Z \rangle + \langle Y, Z \rangle$

(iv) $a, b \in \mathbb{R}$, $\langle aX, bY \rangle = ab \langle X, Y \rangle$

(v) 对于停时 T , $\langle X^T, Y^T \rangle_t = \langle X, Y \rangle_{t \wedge T}$

例 6. B, W 独立布朗运动, 则 $\langle B, W \rangle = 0$, 即 BW 是鞅。

① $\mathbb{E}(B_t W_t | \mathcal{F}_t) = \mathbb{E}(B_t | \mathcal{F}_t) \mathbb{E}(W_t | \mathcal{F}_t) = B_t W_t$

② $B + W, B - W$ 均为布朗运动 $\Rightarrow \langle B, W \rangle = \frac{1}{4} (\langle B + W \rangle - \langle B - W \rangle) = 0$

若 f 是连续有限变差函数, 那么二次变差过程是 0。

$$\forall \Pi: \sum_i |f(t_i) - f(t_{i-1})| \leq \sup |f(u) - f(x)| \cdot |f|_{TV},$$

$$|\Pi| \rightarrow 0 \implies \sup |f(t_i) - f(t_{i-1})| \rightarrow 0.$$

对连续局部鞅的随机积分

$\forall H$ 适应, $\mathbb{E} \int_0^t H_s^2 d\langle M \rangle_s < \infty$, $N_t = \int_0^t H_s dM_s$ well defined, 且有

$$\mathbb{E} \left(\int_0^t H_s dM_s \right)^2 = \mathbb{E} \int_0^t H_s^2 d\langle M \rangle_s,$$

$$\left\langle \int H dM \right\rangle_t = \int_0^t H_s^2 d\langle M \rangle_s.$$

性质:

$$H \cdot (M + N) = H \cdot M + H \cdot N,$$

$$(H + K) \cdot M = H \cdot M + K \cdot M,$$

$$\langle H \cdot M, K \cdot N \rangle = \int_0^t H_s K_s d\langle M, N \rangle_s.$$

Kunita–Watanabe inequality

$$\int_0^t |H_s K_s| d\langle M, N \rangle_s \leq \left(\int_0^t H_s^2 d\langle M \rangle_s \right)^{\frac{1}{2}} \left(\int_0^t K_s^2 d\langle N \rangle_s \right)^{\frac{1}{2}}.$$

4 Itô 公式

Itô 公式 $\iff$ 微积分基本定理。

对于 C^2 的函数 f :

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds,$$

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2} f''(B_t) dt.$$

证明.

$$f(B_t) - f(B_0) = \sum (f(B_{t_i}) - f(B_{t_{i-1}})) = f'(B_{t_{i-1}})(B_{t_i} - B_{t_{i-1}}) + \frac{1}{2} f''(B_{t_{i-1}})(B_{t_i} - B_{t_{i-1}})^2$$

$$\|\Pi\| \rightarrow 0 \implies \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds. \quad \square$$

对于 $f(t, x) \in C^{1,2}([0, \infty) \times \mathbb{R})$,

$$f(t, B_t) - f(0, B_0) = \int_0^t \frac{\partial f}{\partial s}(s, B_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, B_s) dB_s + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(s, B_s) ds,$$

$$df(t, B_t) = f_t dt + f_x dB_t + \frac{1}{2} f_{xx} ds.$$

应用: Heat equation

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}, \quad u(0, x) = \varphi(x).$$

解: 固定 T , $f(t, x) = u(T - t, x)$,

$$\frac{\partial f}{\partial t} = \frac{\partial u(T - t, x)}{\partial t} = -\frac{\partial u}{\partial t}(T - t, x) = -\frac{1}{2} \frac{\partial^2 u}{\partial x^2}(T - t, x) = -\frac{1}{2} \frac{\partial^2 f}{\partial x^2},$$

$$\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} = 0, \quad f(T, x) = \varphi(x).$$

$$\begin{aligned} df(t, B_t) &= f_t(t, B_t) dt + f_x(t, B_t) dB_t + \frac{1}{2} f_{xx}(t, B_t) dt \\ &= f_x(t, B_t) dB_t. \end{aligned}$$

$f(t, B_t) = \int_0^t f_x(s, B_s) dB_s$ 是一个鞅。

$$\mathbb{E}_x(f(0, B_0)) = \mathbb{E}[f(T, B_T)] = f(0, x),$$

$$\implies u(T, x) = \mathbb{E}_x \varphi(B_T).$$

 n 维布朗运动的 Itô 公式

$B_t = (B_t^1, B_t^2, \dots, B_t^n)$ 是 n 维布朗运动, 对于 $f \in C^{1,2}([0, \infty) \times \mathbb{R}^n)$,

$$f(t, B_t) = f(0, B_0) + \int_0^t \frac{\partial f}{\partial s}(s, B_s) ds + \int_0^t \nabla f(s, B_s) \cdot dB_s + \frac{1}{2} \int_0^t \Delta f(s, B_s) ds.$$

Itô 公式:

$$(dB_t^i)^2 = dt, \quad dB_t^i dB_t^j = 0 \ (i \neq j), \quad dt dB_t = 0, \quad dt^2 = 0.$$

连续半鞅

X_t 是关于 $\mathcal{F}_t$ 的连续半鞅是指

$$X_t \sim \mathcal{F}_t, \quad X_t = M_t + A_t,$$

其中 M_t 是局部鞅, A_t 是连续有限变差过程, $A_0 = 0$ 。

注. 这种分解是唯一的。

$$X_t = M_t + A_t = \tilde{M}_t + \tilde{A}_t \implies M_t - \tilde{M}_t = \tilde{A}_t - A_t = k$$

$$\implies \langle k \rangle = 0, \quad k \text{ 是鞅}, \quad k = k_0 = 0.$$

定义: $\langle X_t \rangle = \langle M_t \rangle$ 。

定义是自洽的, 因为有关协方差过程是相同的。

$$\sum_{t_i} (X_{t_{i+1}} - X_{t_i})^2 \xrightarrow{\mathbb{P}} \langle X \rangle_t, \quad \|\Pi\| \rightarrow 0.$$

$$\begin{aligned}
\sum_{t_i} (X_{t_{i+1}} - X_{t_i})^2 &= \sum (M_{t_{i+1}} + A_{t_{i+1}} - M_{t_i} - A_{t_i})^2 \\
&= \sum (M_{t_{i+1}} - M_{t_i})^2 + 2 \sum (M_{t_{i+1}} - M_{t_i})(A_{t_{i+1}} - A_{t_i}) + \sum (A_{t_{i+1}} - A_{t_i})^2 \\
&\quad \downarrow \mathbb{P} \langle X \rangle \quad \downarrow \mathbb{P} 0 \\
&\leq 2 |M_{t_{i+1}} - M_{t_i}| |A_{t_{i+1}} - A_{t_i}| \leq \sqrt{\sum (M_{t_{i+1}} - M_{t_i})^2} \sqrt{\sum (A_{t_{i+1}} - A_{t_i})^2} \xrightarrow{\mathbb{P}} 0 \\
&\quad \int H_s dX_s = \int H_s dM_s + \int H_s dA_s.
\end{aligned}$$

对于 $f \in C_c^2(\mathbb{R})$,

$$f(X_t) - f(X_0) = \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s.$$

对于高维半鞅 $\{X_t^{(k)}, 1 \leq k \leq d\}$ w.r.t. $\{\mathcal{F}_t\}$, 即定义为 $X_t^{(k)} = M_t^{(k)} + A_t^{(k)}$, M 为局部鞅, A 为有限变差过程.

定义协方差过程

$$\langle X^{(i)}, X^{(j)} \rangle_t := \langle M^{(i)}, M^{(j)} \rangle_t.$$

则有 Itô 公式 ($f \in C_c^2(\mathbb{R}^d)$):

$$f(X_t) = f(X_0) + \int_0^t \sum_{i=1}^d \frac{\partial f}{\partial x_i}(X_t) dX_t + \frac{1}{2} \int_0^t \sum_{i,j=1}^d \frac{\partial^2 f}{\partial x_i \partial x_j}(X_t) d\langle X_t^i, X_t^j \rangle.$$

推论: 分部积分公式

$$\begin{aligned}
X_t Y_t &= X_0 Y_0 + \int Y_t dX_t + \int X_t dY_t + \frac{1}{2} \cdot 2 d\langle X_t, Y_t \rangle, \\
\implies X_t Y_t &= X_0 Y_0 + \int Y_t dX_t + \int X_t dY_t + \langle X_t, Y_t \rangle.
\end{aligned}$$

例子: Black-Scholes 模型

$$\frac{dX_t}{X_t} = \mu dt + \sigma dB_t.$$

思路 $\log X_t = \int \mu dt + \int \sigma dB_t$,

$$\begin{aligned}
d \log X_t &= \frac{1}{X_t} dX_t - \frac{1}{2} \frac{1}{X_t^2} d\langle X \rangle_t, \\
dX_t &= \mu X_t dt + \sigma X_t dB_t, \quad X_t = \underbrace{\int \mu X_t dt}_{\text{有界变差}} + \underbrace{\int \sigma X_t dB_t}_{\text{鞅}}, \\
\langle X \rangle_t &= \left\langle \int \sigma X_t dB_t \right\rangle = \int \sigma^2 X_t^2 dt, \quad d\langle X \rangle_t = \sigma^2 X_t^2 dt. \\
\implies d \log X_t &= \mu dt + \sigma dB_t - \frac{1}{2} \sigma^2 dt, \\
\log X_t - \log X_0 &= \left(\mu - \frac{1}{2} \sigma^2\right) t + \sigma B_t, \\
\implies X_t &= X_0 \exp\left(\left(\mu - \frac{1}{2} \sigma^2\right) t + \sigma B_t\right),
\end{aligned}$$

称 X_t 为几何布朗运动。

$$\begin{aligned}
\mu > \frac{\sigma^2}{2} : \lim_t X_t &= \infty \text{ a.s.}; & \mu < \frac{\sigma^2}{2} : \lim_t X_t &= 0 \text{ a.s.}; \\
\mu = \frac{\sigma^2}{2} : X &\text{ 在 } [0, \infty) \text{ 上为鞅.}
\end{aligned}$$

例子：Ornstein–Uhlenbeck process

$$dX_t = dB_t - rX_t dt,$$

$$\begin{aligned} d(e^{rt}X_t) &= rX_te^{rt}dt + e^{rt}dB_t + d\langle e^{rt}, X_t \rangle \\ &= rX_te^{rt}dt + e^{rt}dB_t - re^{rt}X_tdt \\ &= e^{rt}dB_t. \end{aligned}$$

$$e^{rt}X_t - X_0 = \int_0^t e^{rs}dB_s, \quad X_t = e^{-rt}\left(X_0 + \int_0^t e^{rs}dB_s\right).$$

$$e^{-rt} \int_0^t e^{rs}dB_s \xrightarrow{\text{Riemann 和}} \sim N\left(0, \frac{1}{2r}(1 - e^{-2rt})\right),$$

$$\int_0^t e^{rs}ds = \frac{e^{rs}}{r}\Big|_0^t = \frac{e^{rt} - 1}{r}, \quad e^{-2rt} \int_0^t e^{2rs}ds = \frac{1}{2r}(1 - e^{-2rt}).$$

若 $t \rightarrow \infty$, $X_t \approx e^{-rt}X_0 + N(0, \frac{1}{2r})$ 。

Lévy's characterization of Brownian motion

假定 $\{X^{(1)}, \dots, X^{(d)}\}$ 是连续适应过程, 使得 $M_t^{(i)} = X_t^{(i)} - X_0^{(i)}$ 是连续局部鞅, $\langle M^{(i)}, M^{(j)} \rangle_t = \delta_{ij}t$, 那么 $X_t = (X_t^{(1)}, \dots, X_t^{(d)})$ 是 d 维适应于 $\{\mathcal{F}_t\}$ 的布朗运动。

证明. $f(x) = e^{i\xi \cdot x}$, $f(X_t) = e^{i\xi \cdot X_t}$,

$$\frac{\partial f}{\partial x_i} = i\xi_i e^{i\xi \cdot x}, \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = -\xi_i \xi_j e^{i\xi \cdot x},$$

$$df(X_t) = \sum_i \frac{\partial f}{\partial x_i}(X_t) dX_t^i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(X_t) d\langle X^i, X^j \rangle_t,$$

$$f(X_t) - f(X_0) = \sum_i i\xi_i \int_0^t e^{i\xi \cdot X_s} dX_s^i - \sum_i \xi_i^2 \int_0^t e^{i\xi \cdot X_s} ds,$$

$$\forall A \in \mathcal{F}_t, \quad \mathbb{E}(\mathbf{1}_A(f(X_t) - f(X_0))) = \sum_i \mathbb{E} \int_0^t \mathbf{1}_A i\xi_i e^{i\xi \cdot X_s} dX_s^i - \frac{1}{2} |\xi|^2 \mathbb{E} \int_0^t \mathbf{1}_A e^{i\xi \cdot X_s} ds$$

$$\downarrow 0$$

$$\implies \mathbb{E} \mathbf{1}_A f(X_t) = -\frac{1}{2} |\xi|^2 \int_0^t \mathbb{E} \mathbf{1}_A f(X_s) ds,$$

$$\implies \mathbb{E} \mathbf{1}_A f(X_t) = e^{-\frac{|\xi|^2}{2}t} \mathbb{P}(A).$$

因而 X_t 各分量独立, 为高斯分布 $\sim N(0, t)$ 。

对 $X_t - X_s$ 同理, 有平稳增量性和独立增量性。

推论: 任何一个实值连续局部鞅是时间变换下的布朗运动。 □

证明. 假定 Λ_t 满足上述条件,

$$\tau_t = \inf\{s \geq 0 : \langle X \rangle_s \geq t\}, \quad t \geq 0,$$

$$M_t = X_{\tau_t} \implies \langle M \rangle_t = \langle X \rangle_{\tau_t} = t \implies M_t \text{ 为一维布朗运动.}$$

□

5 随机微分方程

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 = \xi.$$

应该理解为积分方程

$$X_t = \xi + \int_0^t b(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dB_s.$$

强解: 给一个 ξ 和 B_t , X_t 在相同的概率空间中构造并满足

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 = \xi.$$

弱解: 可以找到一概率空间, 上面的一个 $y \stackrel{d}{=} \xi$ 和一个布朗运动 W_t , $\tilde{X}_t$ 满足

$$d\tilde{X}_t = b(t, \tilde{X}_t) dt + \sigma(t, \tilde{X}_t) dW_t, \quad \tilde{X}_0 = \xi.$$

定理 7. 令 $T > 0$,

$$b(t, x) : [0, T] \times \mathbb{R}^n \longrightarrow \mathbb{R}^n, \quad \sigma(t, x) : [0, T] \times \mathbb{R}^n \longrightarrow \mathbb{R}^{n \times d},$$

满足

$$|b(t, x)| + |\sigma(t, x)| \leq C(1 + |x|),$$

与

$$|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq C|x - y|.$$

ξ 是一个与 $\mathcal{F}_\infty^\circ = \sigma\{B_s, s \in [0, \infty)\}$ 独立的随机变量, $\mathbb{E}\xi^2 < \infty$, 那么

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 = \xi$$

有唯一连续强解。

反例:

$$\begin{cases} dX_t = X_t^2 dt \\ X_0 = 1 \end{cases} \implies -\frac{1}{X_t} + 1 = t \implies X_t = \frac{1}{1-t},$$

$$\begin{cases} dX_t = 3X_t^{2/3} dt \\ X_0 = 0 \end{cases} \quad \begin{cases} X_t \equiv 0 \\ X_t = t^3 \end{cases}$$

唯一性

若 X, Y 为两解,

$$(X_t - Y_t)^2 = \left(\int_0^t b(s, X_s) ds - \int_0^t b(s, Y_s) ds + \int_0^t \sigma(s, X_s) dB_s - \int_0^t \sigma(s, Y_s) dB_s \right)^2,$$

$$\begin{aligned} \mathbb{E}(X_t - Y_t)^2 &\leq 2\mathbb{E}\left(\int_0^t b(s, X_s) - b(s, Y_s) ds\right)^2 + 2\mathbb{E}\left(\int_0^t (\sigma(s, X_s) - \sigma(s, Y_s)) dB_s\right)^2 \\ &\leq 2\mathbb{E}\left(\int_0^t |b(s, X_s) - b(s, Y_s)| ds\right)^2 + 2\mathbb{E}\int_0^t (\sigma(s, X_s) - \sigma(s, Y_s))^2 ds \\ &\leq 2\mathbb{E} \cdot t \int_0^t (b(s, X_s) - b(s, Y_s))^2 ds + 2\mathbb{E}\int_0^t (\sigma(s, X_s) - \sigma(s, Y_s))^2 ds \\ &\leq 2tC \mathbb{E}\int_0^t (X_s - Y_s)^2 ds + 2C \mathbb{E}\int_0^t (X_s - Y_s)^2 ds. \end{aligned}$$

设 $u_t = \mathbb{E}(X_t - Y_t)^2$, 则

$$\begin{aligned} u_t &\leq (2TC + 2C) \int_0^t u_s \, ds \leq (2TC + 2C) \int_0^t u_s \, ds, \\ \left(e^{-At} u_t - e^{-At} \int_0^t u_s \, ds \right) &\leq 0, \\ \frac{d}{dt} e^{-At} \int_0^t u_s \, ds &\leq 0 \implies e^{-At} \int_0^t u_s \, ds = 0 \implies u_t = 0. \end{aligned}$$

存在性：思路是逐步迭代

$X_t^0 = \xi$, 假定已给出 X_t^n ,

$$X_t^{n+1} = \int_0^t b(s, X_s^n) \, ds + \int_0^t \sigma(s, X_s^n) \, dB_s + \xi.$$

$$\begin{aligned} \mathbb{E}(X_t^n - X_t^{n-1})^2 &= \mathbb{E} \left(\int_0^t b(s, X_s^n) - b(s, X_s^{n-1}) \, ds + \int_0^t \sigma(s, X_s^n) - \sigma(s, X_s^{n-1}) \, dB_s \right)^2 \\ &\leq 2\mathbb{E} \left(\int_0^t b(s, X_s^n) - b(s, X_s^{n-1}) \, ds \right)^2 + 2\mathbb{E} \left(\int_0^t \sigma(s, X_s^n) - \sigma(s, X_s^{n-1}) \, dB_s \right)^2 \\ &\leq 2T\mathbb{E} \int_0^t (b(s, X_s^n) - b(s, X_s^{n-1}))^2 \, ds + 2\mathbb{E} \int_0^t (\sigma(s, X_s^n) - \sigma(s, X_s^{n-1}))^2 \, ds \\ &\leq (2TC + 2C) \int_0^t \mathbb{E}(X_s^n - X_s^{n-1})^2 \, ds \\ \implies \mathbb{E}(X_t^n - X_t^{n-1})^2 &\leq (2TC + 2C)^{n-1} \int_0^t \cdots \int_0^{s_2} \mathbb{E}(X_{s_1} - X_{s_0})^2 \, ds_1, \end{aligned}$$

$$\begin{aligned} \mathbb{E}(X_s^1 - X_s^0)^2 &\leq 2\mathbb{E} \left(\int_0^T b(s, \xi) \, ds \right)^2 + \sigma(s, \xi)^2 \, ds \\ &\leq 2T\mathbb{E} \int_0^T C(1 + |\xi|)^2 \, ds \\ &= 2TC \mathbb{E}(1 + |\xi|)^2 \int_0^T \, ds, \end{aligned}$$

$$\implies \mathbb{E}(X_t^n - X_t^{n-1})^2 \leq \frac{(2TC + 2C)^{n-1} 2TC \mathbb{E}(1 + |\xi|)^2 T^n}{n!}.$$

$$\begin{aligned} \mathbb{P}(|X_t^n - X_t^{n-1}| > 2^{-n}) &\leq 2^{2n} (2TC + 2C)^{n-1} \frac{2TC \mathbb{E}(1 + |\xi|)^2 T^n}{n!} \\ &\leq \sum_n \mathbb{P} \left(\sup_t |X_t^n - X_t^{n-1}| > 2^{-n} \right) < \infty \quad \text{收敛} \end{aligned}$$

$\forall a > 0$, w , $\exists N$, $\forall n > N$, $X_t^n - X_t^{n-1} < 2^{-n} \Rightarrow \sum X_t^n - X_t^{n-1} = X_t^n - \xi$ 收敛。

$\Rightarrow X_t^n$ 几乎必然收敛到一个随机变量, 且由于控制与 t 无关, 关于 t 一致。

显然 $\lim X_t^n$ 满足方程, 且由一致收敛, $\lim X_t^n$ 连续。

6 转移半群与无穷小生成元

$$dX_t = b(X_t) dt + \sigma(X_t) dB_t$$

在上面两个解下, 有强马氏性:

$$\mathbb{E}_x(f(X_{T+t}) \mid \mathcal{F}_T) = \mathbb{E}_{X_T}(f(X_{T+t})).$$

令

$$P_t f(x) := \mathbb{E}_x[f(X_t)], \quad P_0 f = f.$$

对于 $t, s > 0$,

$$\begin{aligned} P_{t+s} f(x) &= \mathbb{E}_x f(X_{t+s}) = \mathbb{E}_x \left(\mathbb{E}(f(X_{t+s}) \mid \mathcal{F}_s) \right) = \mathbb{E}_x \mathbb{E}_{X_s} f(X_{t+s}) \\ &= \mathbb{E}_x (P_t f(X_s)) = P_s P_t f(x), \end{aligned}$$

具有半群的性质。

无穷小生成元

$$L f(x) = \frac{d}{dt} P_t f(x) \Big|_{t=0} = \lim_{\delta \rightarrow 0} \frac{P_\delta f(x) - f(x)}{\delta},$$

L 的定义域为 $\text{Dom}(L) = \{f : \text{上述极限存在}\}$ 。

当 $t > 0$,

$$\begin{aligned} \frac{d}{dt} P_t f(x) &= \lim_{\delta \rightarrow 0} \frac{P_{t+\delta} f(x) - P_t f(x)}{\delta} = \frac{P_t P_\delta f(x) - P_t f(x)}{\delta} = L P_t f(x), \\ \frac{d}{dt} P_t f(x) &= \lim_{\delta \rightarrow 0} \frac{P_{t+\delta} f(x) - P_t f(x)}{\delta} = P_t \lim_{\delta \rightarrow 0} \frac{P_\delta f(x) - f(x)}{\delta} = P_t L f(x). \end{aligned}$$

第一个是科尔莫戈洛夫向后方程, 第二个是福克-普朗克方程。

对于 $f \in C_b^2(\mathbb{R})$, $P_t f(x) = \mathbb{E}_x f(X_t)$,

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s \\ &= \int_0^t f'(X_s) b(X_s) ds + \int_0^t f'(X_s) \sigma(X_s) dB_s + \frac{1}{2} \int_0^t f''(X_s) \sigma(X_s)^2 ds, \end{aligned}$$

$$\begin{aligned} P_t f(x) - f(x) &= \mathbb{E}_x (f(X_t) - f(X_0)) \\ &= \mathbb{E}_x \int_0^t f'(X_s) b(X_s) ds + \mathbb{E}_x \int_0^t f'(X_s) \sigma(X_s) dB_s + \frac{1}{2} \mathbb{E}_x \int_0^t f''(X_s) \sigma^2(X_s) ds \\ &= \mathbb{E}_x \int_0^t f'(X_s) b(X_s) ds + \frac{1}{2} \mathbb{E}_x \int_0^t f''(X_s) \sigma^2(X_s) ds, \\ &\implies L f(x) = f'(x) b(x) + \frac{1}{2} f''(x) \sigma^2(x), \end{aligned}$$

所以 $C_b^2 \subset \text{Dom}(L)$ 。

对于高维形式 $dX_t = b(X_t) dt + \sigma(X_t) dB_s$,

$$L f(x) = \frac{1}{2} \sum_{i,j} a_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x) + b(x) \cdot \nabla f(x),$$

其中 $(a_{ij}) = \sigma(x) \sigma(x)^T$ 。

$$P_t f(x) = \mathbb{E}_x(f(X_t)),$$

$$\begin{aligned} f(X_t) - f(x) &= \nabla f(X_t) \cdot dX_t + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(X_t) d\langle X_t^i, X_t^j \rangle \\ &= \nabla f(X_t) \sigma(X_t) dB_s + \nabla f(X_t) b(X_t) dX_t + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} a_{ij} ds, \end{aligned}$$

$$dX_t^i = b^i(X_t) dt + \sigma^{i\cdot}(X_t) dB_s,$$

$$dX_t^j = b^j(X_t) dt + \sigma^{j\cdot}(X_t) dB_s$$

$$\implies d\langle X^i, X^j \rangle = \sigma^{i\cdot}(X_t) \sigma^{j\cdot}(X_t) ds \implies (a_{ij}) = \sigma \cdot \sigma^T,$$

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{P_t f(x) - f(x)}{t} &= \mathbb{E} \nabla f(X_t) \sigma(X_t) dB_s - \nabla f(X_t) b(X_t) dX_t + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} a_{ij} \\ &= \nabla f(x) \cdot b(x) + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(x) a_{ij}(x). \end{aligned}$$

7 Kolmogorov 方程

Kolmogorov 向后方程

$$\frac{\partial}{\partial t} P_t f(x) = L P_t f(x).$$

若 X_t 有密度 $p(t, x, y)$ (从 x 出发, 过时间 t 后到达 y 的密度函数),

$$P_t f(x) = \mathbb{E}_x f(X_t) = \int f(y) p(t, x, y) dy,$$

$$\int f(y) \frac{\partial}{\partial t} p(t, x, y) dy = \int f(y) L p(t, x, y) dy,$$

$$\frac{\partial}{\partial t} p(t, x, y) = L p(t, x, y) \quad \text{a.s.}$$

Fokker-Planck equation

$$\frac{d}{dt} P_t f(x) = P_t L f,$$

$$\frac{d}{dt} \int_{\mathbb{R}^n} p(t, x, y) f(y) dy = \int_{\mathbb{R}^n} p(t, x, y) L f(y) dy = \int_{\mathbb{R}^n} L^* p(t, x, y) f(y) dy,$$

L^* 为对偶算子: $\varphi \in C_c^2(\mathbb{R}^n)$,

$$L^* \varphi(y) = \sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} (a_{ij}(y) \varphi(y)) - \sum_j \frac{\partial}{\partial x_j} (b_j(y) \varphi(y)),$$

$$\implies \frac{\partial}{\partial t} p(t, x, y) = L_y^* p(t, x, y).$$

μ_t 代表 X_t 的分布, $\mu_t(dy) = \mu_t(y) dy$,

$$\mu_t(y) = \int_{\mathbb{R}^n} p(t, x, y) \mu_0(dx),$$

$$\implies \frac{\partial}{\partial t} \mu_t(y) = L_y^* \mu_t(y).$$

8 Girsanov transform

$$\mathbb{Q} \ll \mathbb{P}, \quad M_t = \frac{d\mathbb{Q}}{d\mathbb{P}},$$

$\{X_t\}$ 是一个 $\mathbb{Q}$ -鞅 $\iff \{M_t X_t\}$ 是一个 $\mathbb{P}$ -鞅。

$$\mathbb{E}^{\mathbb{Q}}(X_t | \mathcal{F}_s) = X_s \iff \mathbb{E}^{\mathbb{Q}}(\mathbf{1}_A X_t) = \mathbb{E}^{\mathbb{Q}}(\mathbf{1}_A X_s) \quad \forall A \in \mathcal{F}_s$$

$$\iff \int \mathbf{1}_A X_t M_t d\mathbb{P} = \int \mathbf{1}_A X_s M_s d\mathbb{P} \iff M_t X_t \text{ 是鞅.}$$

定理 8. 假定 $\{B_t\}$ 是一个布朗运动, $\{\xi_t, t \geq 0\}$ 是适应过程,

$$\mathbb{E} \exp\left(\frac{1}{2} \int_0^T |\xi_s|^2 ds\right) < \infty,$$

$$M_t = \exp\left\{\int_0^t \xi_s dB_s - \frac{1}{2} \int_0^t |\xi_s|^2 ds\right\}$$

是鞅。

证明. $dM_t = M_t \xi_t dB_t$ 。

令 $X_t = \int_0^t \xi_s dB_s - \frac{1}{2} \int_0^t |\xi_s|^2 ds$, $dM_t = M_t \xi_t dB_t \Rightarrow M_t$ 是局部鞅, 用 Novikov 条件, M_t 是鞅。 $\square$

定理 9 (Girsanov).

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_t = \exp\left\{\int_0^t \xi_s dB_s - \frac{1}{2} \int_0^t |\xi_s|^2 ds\right\},$$

$$\tilde{B}_t := B_t - \int_0^t \xi_s ds$$

是在 $\mathbb{Q}$ 下的布朗运动。

证明.

$$\begin{aligned} d(\tilde{B}_t M_t) &= M_t d\tilde{B}_t + \tilde{B}_t dM_t + d\langle M, \tilde{B} \rangle_t \\ &= M_t (dB_t - \xi_t ds) + \tilde{B}_t dM_t + M_t \xi_t dt \\ &= M_t dB_t + \tilde{B}_t dM_t, \end{aligned}$$

$\Rightarrow M_t \tilde{B}_t$ 是鞅。

$$\text{在 } \mathbb{P} \text{ 下 } \langle \tilde{B}^i, \tilde{B}^j \rangle_t = \langle B^i, B^j \rangle_t = \begin{cases} t & i = j \\ 0 & i \neq j \end{cases},$$

在 $\mathbb{Q}$ 下 $\langle \tilde{B}^i, \tilde{B}^j \rangle_t = \delta_{ij} t$, 故 $\mathbb{Q} \ll \mathbb{P}$, $\Rightarrow \tilde{B}$ 在 $\mathbb{Q}$ 上是布朗运动。 $\square$

应用: 假定 $b(t, x) : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ 有界,

$$dX_t = dB_t + b(t, X_t) dt, \quad X_0 = x$$

有一个唯一弱解。

证明 (存在性). ① 存在性: $\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp\left\{\int_0^t b(s, X_s) dX_s - \frac{1}{2} \int_0^t |b(s, X_s)|^2 ds\right\}$, X_t 为 $\mathbb{P}$ 上的布朗运动,

$$\Rightarrow \tilde{B}_t := X_t - x - \int_0^t b(s, X_s) ds \text{ 为 } \mathbb{Q} \text{ 上的布朗运动,}$$

X_t 在 $\mathbb{Q}$ 上解上述方程。

② 唯一性:

$$dX_t = dB_t + b(t, X_t) dt,$$

令

$$N_t = \exp\left\{-\int_0^t b(s, X_s) dB_s - \frac{1}{2} \int_0^t |b(s, X_s)|^2 ds\right\}, \quad \frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_t} = N_t,$$

$$X_t - x = B_t + \int_0^t b(s, X_s) ds$$

是 $\mathbb{Q}$ 上的布朗运动, $\Rightarrow X_t$ 在 $\mathbb{Q}$ 上按分布唯一确定 $\Rightarrow X_t$ 唯一。

□

9 SDE 与 PDE

回顾 n 维 SDE

$$dX_t = \sigma(X_t) dB_t + b(X_t) dt,$$

$$\sigma(x) = (\sigma_{ij}(x))_{n \times n}, \quad b(x) = \begin{pmatrix} b_1(x) \\ \vdots \\ b_n(x) \end{pmatrix}, \quad X_t = \begin{pmatrix} X_t^1 \\ \vdots \\ X_t^n \end{pmatrix},$$

分量上

$$dX_t^i = \sum_{j=1}^n \sigma_{ij}(X_t) dB_t^j + b_i(X_t) dt,$$

$$d\langle X^i, X^j \rangle_t = dX_t^i dX_t^j = \sum_{k=1}^n \sigma_{ik}(X_t) \sigma_{jk}(X_t) dt = a_{ij}(X_t) dt,$$

$$A(t, x) := (a_{ij}(x)) = \sigma(x) \sigma(x)^T.$$

考虑 $f(t, x) \in C^{1,2}([0, \infty) \times \mathbb{R}^n)$,

$$f(t, X_t) = f(0, X_0) + \text{鞅部分} + \int_0^t \left(\frac{\partial}{\partial s} + L \right) f(s, X_s) ds,$$

其中

$$L\psi(x) = \frac{1}{2} \sum_{i,j=1}^n a_{ij}(x) \frac{\partial^2 \psi(x)}{\partial x_i \partial x_j} + b(x) \cdot \nabla \psi(x).$$

定理 10. 假定 $u = u(t, x) \in C^{1,2}([0, \infty) \times \mathbb{R})$ 是

$$\frac{\partial u}{\partial t} = Lu(t, x), \quad u(0, x) = \varphi(x)$$

的解, 那么 $u(t, x) = \mathbb{E}_x[\varphi(X_t)]$ 。

证明. 固定 T , $f(t, x) = u(T - t, x)$,

$$\Rightarrow \frac{\partial f}{\partial t} + Lf = 0 \Rightarrow f(t, X_t) \text{ 是鞅,}$$

$$\mathbb{E}_x f(0, X_0) = \mathbb{E}_x f(T, X_T), \quad u(T, x) = \mathbb{E}_x \varphi(X_T).$$

□

Black-Scholes 模型

$$dS_t = S_t(\mu_t dt + \sigma_t dB_t),$$

r_t 为利率 (银行), μ_t 为增长率, $\tilde{\mu}_t = \mu_t - r_t$ 为真正的利率, σ_t 为扰动项。

欧氏期权: 持有者在 T 时有以 K 的价格买入 S 的权利, 最终收益为 $(S_T - K)^+$ 。

问题是如何定价。方法是构造一个投资策略 $(w_0(t), w_1(t), \dots, w_d(t))$, 使得其价值在 T 时刻恰好是 $(S_T - K)^+$, 构建投资策略所需的起始资金为合理定价。

将最终收益抽象为 $\psi(S_T)$ 。

为了使得货币价格有可比性, 我们引入折现过程

$$dA_t = r(t)A_t dt, \quad A_t = e^{\int r(s) ds}.$$

期权行为

$$dS_t^{(i)} = S_t^{(i)} \left(r(t) dt + \tilde{\mu}_i(t) dt + \sum_{j=1}^d \sigma_{ij}(t) dB_t^{(j)} \right),$$

假定全部都有界。

$$\begin{aligned} d(A_t^{-1} S_t^{(i)}) &= S_t^{(i)} dA_t^{-1} + A_t^{-1} dS_t^{(i)} \\ &= S_t^{(i)} A_t^{-1} (-r(t) dt) + S_t^{(i)} A_t^{-1} \left(r(t) dt + \tilde{\mu}_i(t) dt + \sum_j \sigma_{ij}(t) dB_t^{(j)} \right) \\ &= A_t^{-1} S_t^{(i)} \sum_j \sigma_{ij}(t) d\tilde{B}_t^{(j)}, \end{aligned}$$

$$\tilde{B}_t^{(j)} = B_t^{(j)} + \int_0^t \sigma^{jk}(s) \tilde{\mu}_k(s) ds.$$

投资组合 $(w_0(t), w_1(t), \dots, w_d(t))$,

财富过程

$$X_t = \sum_{i=0}^d w_i(t) S_t^{(i)}, \quad S_t^{(0)} := A_t.$$

自融资过程

$$d(A_t^{-1} X_t) = \sum_{i,j=1}^d w_i(t) A_t^{-1} S_t^{(i)} \sigma_{ij}(t) d\tilde{B}_t^{(j)}.$$

目标: 找 $(w_0(t), w_1(t), \dots, w_d(t))$ 使得 $X_T = \psi(S_T)$ 。

定理 11 (对布朗运动的鞅表示定理). 对于 $\xi \in L^2(\mathcal{F}_T^+)$, $\exists \mathcal{F}_s^+$ 可测过程 H_s , $\mathbb{E} \int_0^T |H_s|^2 ds < \infty$,

$$\xi = \mathbb{E}\xi + \int_0^T H_s dB_s.$$

H_s 在下面的意义下唯一:

$$\mathbb{E} \int_0^T |H_s - H'_s|^2 ds = 0.$$

令

$$b_j(s) = \sum_{k=1}^d \sigma^{jk}(s) \tilde{\mu}_k(s).$$

定理 12 (Risk-neutral measure $\mathbb{Q}$).

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\sum_{j=1}^d \int_0^t b_j(s) dB_s^j - \frac{1}{2} \sum_{j=1}^d \int_0^t (b_j(s))^2 ds\right),$$

在 $\mathbb{Q}$ 的意义下

$$\bar{B}_t = B_t + \int_0^t b(s) ds$$

是一个 BM。

所以在 $\mathbb{Q}$ 下

$$d(A_t^{-1}X_t) = \sum_{i,j} w_i(t) A_t^{-1} S_t^{(i)} \sigma_{ij}(t) d\bar{B}_t^{(j)},$$

$$A_t^{-1}X_t = \mathbb{E}^{\mathbb{Q}}(A_T^{-1}X_T \mid \mathcal{F}_t^+) = \mathbb{E}^{\mathbb{Q}}(A_T^{-1}\psi(S_T) \mid \mathcal{F}_t^+),$$

$$X_0 = \mathbb{E}^{\mathbb{Q}}A_T^{-1}\psi(S_T).$$

考虑 $d = 1$, $dS_t = S_t(\mu dt + \sigma dB_t)$,

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\sigma^{-1}(\mu - r)B_t - \frac{1}{2}\sigma^{-2}(\mu - r)^2t\right),$$

在 $\mathbb{Q}$ 下 $\tilde{B}_t = B_t + \sigma^{-1}(\mu - r)t$ 是一个布朗运动,

$$X_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}(\psi(S_T) \mid \mathcal{F}_t^+) = e^{-r(T-t)}\mathbb{E}_{S_t}^{\mathbb{Q}}(\psi(S_{T-t})) = e^{-r(T-t)}u(T-t, S_t),$$

$$u(t, x) = \mathbb{E}_x^{\mathbb{Q}}(\psi(S_T)),$$

进一步

$$X_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\left(\psi\left(S_t \exp((T-t)(r - \frac{\sigma^2}{2}) + \sigma(\bar{B}_T - \bar{B}_t))\right) \mid S_t\right) = v(t, x),$$

$$\implies \text{宏观 BS 方程 } \frac{\partial u}{\partial t} = Lu.$$

10 高维 SDE 与 Itô 公式; Fokker-Planck 的再推导

$f \in C_b^2(\mathbb{R}^d)$, $t \geq 0$,

$$Lf(x) = \frac{1}{2} \sum_{i,j=1}^d a_{ij}(t, x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{i=1}^d b_i(t, x) \frac{\partial f}{\partial x_i}(x).$$

由 Itô 公式

$$g(t, X_t) = g(0, X_0) + \int_0^t \nabla g(s, X_s) \sigma(s, X_s) dB_s + \int_0^t (\partial_s + L)g(s, X_s) ds.$$

令 $g(t, x) = f(x)$, 求期望

$$\implies \mathbb{E}(f(X_t)) = \mathbb{E}f(X_0) + \mathbb{E} \int_0^t Lf(X_s) ds = \mathbb{E}f(X_0) + \int_0^t \mathbb{E}Lf(X_s) ds.$$

μ_t 代表 X_t 分布,

$$\int_{\mathbb{R}^d} f(y) \mu_t(dy) = \int_{\mathbb{R}^d} f(y) \mu_0(dy) + \int_0^t \int_{\mathbb{R}^d} Lf(y) \mu_s(dy) ds,$$

$$\begin{aligned}
\langle f, \mu \rangle &= \int_{\mathbb{R}^d} f(y) \mu(dy), \\
\implies \langle f, \mu_t \rangle &= \langle f, \mu_0 \rangle + \int_0^t \langle Lf, \mu_s \rangle ds, \\
\frac{d}{dt} \langle f, \mu_t \rangle &= \langle Lf, \mu_t \rangle = \langle f, L^* \mu \rangle \quad (L^* \text{ 为对偶}),
\end{aligned}$$

L^* 为对偶算子,

$$\implies \frac{d}{dt} \mu_t = L_t^* \mu_t.$$

若 $\sigma(t, x) = I_{n \times n}$,

$$\begin{aligned}
dX_t &= dB_t + b(t, X_t) dt, \\
L_t f &= \frac{1}{2} \Delta f + b(t, x) \cdot \nabla f, \quad L^* g = \frac{1}{2} \Delta g - \operatorname{div}(b(t, x)g),
\end{aligned}$$

若 $\mu_t(dx) = p(t, x) dx$,

$$\begin{aligned}
\int_{\mathbb{R}^d} f(x) p(t, x) dx &= \int_{\mathbb{R}^d} f(x) \mu_0(dx) + \int_0^t \int_{\mathbb{R}^d} L_s f(y) p(s, y) dy ds \\
&= \int_{\mathbb{R}^d} f(x) \mu_0(dx) + \int_0^t \int_{\mathbb{R}^d} f(x) L_s^* p(s, x) dx ds, \\
\frac{\partial}{\partial t} p(t, x) &= L_t^* p(t, x) = \frac{1}{2} \Delta p(t, x) - \operatorname{div}(p(t, x)b(t, x)).
\end{aligned}$$

时间倒转

$$\begin{aligned}
v_t &= \mu_{T-t}, \quad g(t, x) = p(T-t, x), \\
\frac{\partial g}{\partial t} &= -\frac{\partial p}{\partial t}(T-t, x) = -\left(\frac{1}{2} \Delta p(T-t, x) - \operatorname{div}(p(T-t, x)b(T-t, x))\right) \\
&= -\frac{1}{2} \Delta p(T-t, x) + \operatorname{div}(p(T-t, x)b(T-t, x)) \\
&= \frac{1}{2} \Delta g(t, x) - \Delta p(T-t, x) + \operatorname{div}(g(T-t, x)b(T-t, x)) \\
\Delta p(T-t, x) &= \operatorname{div}(\nabla p(T-t, x)) \\
&= \operatorname{div}(p(T-t, x) \nabla \log p(T-t, x)) \\
&= \operatorname{div}(g(t, x) \nabla \log p(T-t, x)), \\
\frac{\partial g(t, x)}{\partial t} &= \frac{1}{2} \Delta g(t, x) - \operatorname{div}\left(\underbrace{(\nabla \log p(T-t, x) - b(T-t, x))}_{\tilde{b}(t, x)} g(t, x)\right)
\end{aligned}$$

$$\implies g(t, x) \text{ 是 } Y_t \text{ 的密度}, \quad dY_t = dW_t + \tilde{b}(t, Y_t) dt, \quad Y_0 \stackrel{d}{=} X_T.$$

$\nabla \log p(T-t, x)$ 叫 Score function。

Ornstein-Uhlenbeck process

$$dX_t = dB_t - \lambda X_t dt \implies X_t = e^{-\lambda t} X_0 + \int_0^t e^{-\lambda(t-s)} dB_s.$$

对于 OU 过程 $b(x) = -\lambda x$,

$$\tilde{b}(t, x) = \nabla \log p(T-t, x) + \lambda x.$$

T 足够大, $X_T \stackrel{d}{\approx} N(0, \frac{1}{2\lambda})$,

$$dY_t = dW_t + \tilde{b}(t, Y_t) dt, \quad Y_0 \stackrel{d}{=} X_T \approx N(0, \frac{1}{2\lambda}),$$

则 $Y_T \stackrel{d}{=} X_0$ 。

11 Feynman-Kac formula

$$dX_t = \sigma(X_t) dB_t + b(X_t) dt,$$

$q(x)$ 为 $\mathbb{R}^n$ 上有界函数, 如何解

$$\frac{\partial}{\partial t} u(t, x) = Lu(t, x) + q(x)u(t, x), \quad u(0, x) = \varphi(x)?$$

定义

$$e_q(t) := \exp\left(\int_0^t q(s) ds\right).$$

定理 13. 假定 $u \in C^{1,2}([0, \infty) \times \mathbb{R}^n)$ 是上面的解,

$$u(t, x) = \mathbb{E}_x(e_q(t)\varphi(X_t)).$$

证明. $T > 0$, $Y_t = u(T - t, X_t)$, $Z_t = e_q(t)$,

$$dY_t = -\partial_t u(T - t, X_t) dt + d \text{ martingale} + Lu(T - t, X_t) dt,$$

$$dZ_t = q(X_t)e_q(t) dt,$$

$$\begin{aligned} d(Y_t Z_t) &= Y_t dZ_t + Z_t dY_t \\ &= e_q(t)(-\partial_t u + Lu + qu)(T - t, X_t) dt + d \text{ martingale} \\ &= d \text{ martingale}. \end{aligned}$$

□

Dirichlet boundary value problem

假定 $D \subset \mathbb{R}^n$ 是一个光滑区域, $h \in C^2(D) \cap C(\bar{D})$ 是

$$\Delta h(x) = 0, \quad h = \varphi \text{ on } \partial D$$

那么

$$h(x) = \mathbb{E}_x[\varphi(B_{T_D})],$$

T_D 是布朗运动第一次到达边界的时间。

若

$$(L + q)h(x) = 0 \text{ in } D, \quad h = \varphi \text{ on } \partial D,$$

$$h(x) = \mathbb{E}_x(e_q(T_D)\varphi(X_{T_D})).$$